

C.Route

Data Dictionary — BigQuery Dataset: croute_market

Version 1.0 · 21 Aug 2026 · croute-hackathon

Overview

C.Route stores all career market intelligence in a single BigQuery dataset (**croute_market**) inside the GCP project **croute-hackathon**. The dataset has **5 tables** and **3 analytical views**. The tables are the source of truth. The views are pre-computed abstractions that the Python backend queries — never the raw tables directly.

Core principle: **Data decides. AI explains.** Every number shown in the C.Route UI originates from these tables. Gemini receives pre-computed scores and explains them — it never generates market statistics.

Data Flow

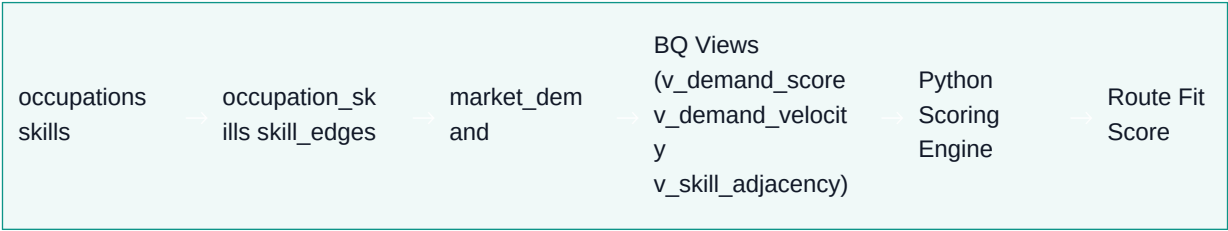


Table Reference

■ occupations

Purpose

The master list of career destinations C.Route can recommend. Every route leads to an occupation.

Connects to

occupation_skills (via occupation_id)
market_demand (via occupation_id)

Field	Type	Description
occupation_id	STRING	Unique identifier for the occupation. Primary key. <i>e.g. OCC001</i>
title	STRING	Human-readable job title shown in the UI and demo. <i>e.g. Business Analyst</i>
category	STRING	Domain grouping — used for filtering and display. Values: analytics, finance, product, marketing. <i>e.g. analytics</i>
geography	STRING	Market geography this occupation applies to. Enables multi-market expansion. <i>e.g. India</i>

Why this table exists: Every other table references occupation_id. This is the anchor table — it defines what careers exist in the system.

Demo relevance: Aisha sees 5 career cards on Screen 3, each sourced from this table.

■ skills

Purpose

The master list of capabilities — both technical and soft. Every skill gap and route score is computed using these.

Connects to

occupation_skills (via skill_id) market_demand
(via skill_id) skill_edges (via skill_a, skill_b)

Field	Type	Description
skill_id	STRING	Unique identifier for the skill. Primary key. <i>e.g. SKL001</i>
skill_name	STRING	Human-readable skill name shown in the UI. <i>e.g. SQL</i>
category	STRING	Type of skill. Values: technical, soft, domain. Used to group skill gaps by type. <i>e.g. technical</i>

Why this table exists: Decouples skill names from all other tables. If a skill name changes, only this table updates — nothing else breaks.

Demo relevance: Skill gap screen shows Aisha's missing skills by name, sourced from this table.

■ occupation_skills

Purpose

Maps which skills each occupation needs and how important each skill is. The bridge between roles and capabilities.

Connects to

occupations (via `occupation_id`) skills (via `skill_id`)

Field	Type	Description
occupation_id	STRING	Foreign key to occupations table. <i>e.g. OCC001</i>
skill_id	STRING	Foreign key to skills table. <i>e.g. SKL001</i>
importance	FLOAT64	How important this skill is for this occupation. Range: 0.0 to 1.0. Used directly in skill gap ranking. <i>e.g. 0.90</i>
period	STRING	The time period this mapping applies to. Allows importance to evolve over time. <i>e.g. 2026</i>

Why this table exists: This table powers two critical calculations: *skill_overlap* (how many required skills does the user have?) and *gap_effort* (how important are the skills they are missing?).

Demo relevance: When Aisha's skills are compared to Business Analyst, this table determines the 67% overlap score.

■ market_demand

Purpose

The market intelligence table. Stores how much demand exists for each skill within each occupation, across time periods. This is what makes C.Route data-driven.

Connects to

occupations (via `occupation_id`) skills (via `skill_id`) Powers all 3 BQ views

Field	Type	Description
<code>occupation_id</code>	STRING	Foreign key to occupations table. <i>e.g. OCC002</i>
<code>skill_id</code>	STRING	Foreign key to skills table. <i>e.g. SKL004</i>
<code>period</code>	STRING	Time period. Multiple periods enable velocity calculation. Current: 2025 (prev), 2026 (current). <i>e.g. 2026</i>
<code>demand_count</code>	INT64	Number of job postings mentioning this skill for this occupation in this period. <i>e.g. 490</i>
<code>demand_share</code>	FLOAT64	Proportion of job postings for this occupation that mention this skill. Range: 0.0 to 1.0. <i>e.g. 0.88</i>
<code>source</code>	STRING	Data source label. Currently <code>synthetic_v1</code> . In production: <code>linkedin</code> , <code>naukri</code> , etc. <i>e.g. synthetic_v1</i>

Why this table exists: Without this table, C.Route has no market grounding. It is the difference between 'Gemini's opinion' and 'what the market actually demands'. Two periods enable velocity — you cannot say a skill is growing without before and after numbers.

Demo relevance: The Career Radar panel and all Route Fit scores originate from this table. When a judge asks 'where does the data come from?' — this is the answer.

■ skill_edges

Purpose

A graph of skill relationships — which skills co-occur together in job postings. Powers the adjacency factor in the Route Fit formula and future learning path recommendations.

Connects to

skills (via skill_a and skill_b)

Field	Type	Description
skill_a	STRING	First skill in the co-occurrence pair. Foreign key to skills. <i>e.g. SKL001</i>
skill_b	STRING	Second skill in the co-occurrence pair. Foreign key to skills. <i>e.g. SKL004</i>
cooccurrence	FLOAT64	Strength of the co-occurrence relationship. Range: 0.0 to 1.0. Higher = more frequently appear together. <i>e.g. 0.85</i>
period	STRING	Time period. Allows adjacency relationships to evolve as the market changes. <i>e.g. 2026</i>

Why this table exists: If Aisha already knows SQL and we know SQL and Python almost always appear together (cooccurrence=0.85), Python is an easier next skill for her than one with no adjacency. This makes learning path recommendations smarter and more personalised.

Demo relevance: Powers the adjacency factor (10%) in the Route Fit formula. Also informs the 90-day roadmap skill sequencing.

BigQuery Views

These 3 views are pre-computed analytical abstractions built on top of the raw tables. The Python backend always queries views — never raw tables. All views use **MAX(period)** dynamically — adding new period data automatically updates results.

View	Built from	What it answers
v_demand_score	market_demand + occupations	How in-demand is each occupation right now? Returns total_demand and avg_demand_share per occupation.
v_demand_velocity	market_demand + occupations + skills	Which occupations and skills are growing? Returns velocity_pct = growth rate current vs previous period.
v_skill_adjacency	skill_edges + skills	Which skills travel together? Returns co-occurrence strength with human-readable skill names.

Route Fit Formula

The Route Fit score is the core output of C.Route. It is computed entirely from BigQuery data — not by Gemini.

Factor	Weight	Source table / view	What it measures
skill_overlap	40%	occupation_skills	Fraction of required skills the user already has
market_demand	25%	v_demand_score	How in-demand this occupation is right now
demand_velocity	15%	v_demand_velocity	Growth rate of this occupation's skills
skill_adjacency	10%	v_skill_adjacency	How close existing skills are to required skills
gap_effort (penalty)	-10%	occupation_skills	How much effort is needed to close the skill gap

Note: skill_adjacency and gap_effort are implemented in the Python scoring engine (Phase 2). The SQL prototype uses the first 3 factors (80% of total weight) to validate the concept.